VIETNAM NATIONAL UNIVERSITY, HO CHI MINH CITY
HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY
OFFICE FOR INTERNATIONAL STUDY PROGRAM
FACULTY OF ELECTRICAL AND ELECTRONICS ENGINEERING

*

TITLE
Subtitle

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Ho Chi Minh City, July 13, 2024

Acknowledgements

Write the Acknowledgment here.

Ho Chi Minh City, July 13, 2024

Signature

Abstract

Write the abstract of your thesis here.

Contents

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1. Introduction

1.1 Motivation

The motivation section should address the following points:

- The significance of your research topic.
- The problem or gap in knowledge that your research addresses.
- Why this problem is worth investigating?.
- Any potential applications or impacts of your research.

1.2 Related Works

In the related works section, you should:

- Review existing literature and research relevant to your topic.
- Discuss the findings and methodologies of previous studies.
- Identify the gaps or limitations in existing research that your thesis aims to address.

Fig. 1.1 illustrates an example Image in the report. The Image should have a caption, and the caption must be placed after the Image. The "label" for figures should begin with "fig:name" and the name should be related to the content of the figure.

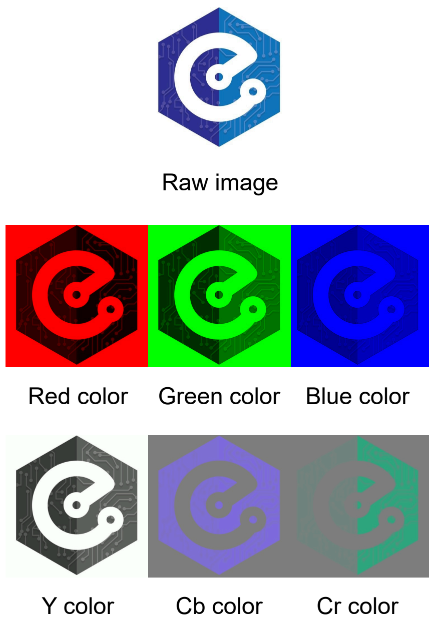

Figure 1.1: The logo of the Department of Electronics in RGB and YCbCr color spaces

1.3 Thesis Contributions

This section should outline the specific contributions of your thesis, such as:

- New theories, models, or frameworks developed.
- Novel methodologies or approaches proposed.
- Significant findings or insights.
- Practical implementations or applications.

1.4 Thesis Structure

Provide a brief overview of the structure of your thesis, summarizing the content and purpose of each chapter.

2. Preliminary

This chapter should cover foundational concepts and background information necessary for understanding your research. Include:

- Basic theories and principles.
- Key terminology and definitions.
- Relevant technologies and tools.
- Any assumptions or limitations.

Anything you want to cite should be cited with the "cite" command. For example, This is a sentence from [1] (A Website) or this is a sentence from [2] (A paper). You can put the citing bib in the file "refs.bib", you can write the bib information yourself, get it from Google Scholar, or use Zotero (free) the best bib library software. The cite name is usually automatically generated, but you can change it to some name you remember, but don't number it.

Table 2.1 is an example table in the report. The Table also needs a caption, and the caption must be placed before the Table. The "label" for table should begin with "tab:name" and the name should be related to the content of the table. You could try to ask ChatGPT combine data to a LaTeX table for you or use `https://www.tablesgenerator.com`. Here' another basic table, Table 2.2.

Table 2.1: JPEG specification defined compression process

Baseline	**Lossless**
- DCT-based process - 8-bit samples - Sequential - Huffman coding with up to 2 AC tables and 2 DC tables - Up to 4 components	- Predictive process - Between 2-bit and 16-bit samples - Huffman or Arithmetic coding with up to 4 AC tables and 4 DC tables - Up to 4 components
Extended DCT based	**Hierarchical**
- DCT-based process - 8-bit or 12-bit samples - Sequential or Progressive - Huffman or Arithmetic coding with up to 4 AC tables and 4 DC tables - Up to 4 components	- Extended DCT-based or lossless process - Multiple Frames - Up to 4 components

Table 2.2: Example Table

Column 1	**Column 2**	**Column 3**
Data 1	Data 2	Data 3
Data 4	Data 5	Data 6
Data 7	Data 8	Data 9

Pay attention to equation, it should use a "begin{equation}" and use its number. Then label it with "eq:name" a name that suggest to you the content. For example, see Equation 2.1.

$$1 + 1 = 2 \tag{2.1}$$

Other than that, small math equation, note, ... should be written in dollar sign "$$" such as $a = b + c$. You can also write algorithm. Here is a basic example of pseudocode using the algorithm2e package, Algorithm 1:

```
Data: Input data
Result: Output result
Initialization;
while condition do
    perform action;
    if condition then
        perform action 1;
    else
        perform action 2;
    end
end
```

Algorithm 1: Example Algorithm

3. Software Implementation

Detail the software development aspect of your project. Discuss:

- The programming languages, tools, and frameworks used.
- The design and architecture of your software solution.
- Key algorithms and data structures.
- Any challenges faced and how they were addressed.
- Testing and validation procedures.

4. Hardware Implementation

4.1 Hardware Design

In this section, cover the hardware design aspects, including:

- The choice of FPGA platform.
- Design specifications and requirements.
- The hardware architecture and components.
- The design process, including schematic and layout considerations.
- Any simulations or analyses performed to validate the design.

4.2 Software Design

Discuss the software design that complements the hardware implementation. Include:

- The development environment and tools used.
- The integration between hardware and software.
- Firmware or drivers developed for the FPGA.
- Software algorithms and their implementation.
- Testing and debugging procedures.

5. Hardware Simulation

This chapter should focus on the simulation aspect of your hardware design. Include:

- The simulation tools and environments used.
- Simulation models and parameters.
- The process of running simulations and analyzing results.
- Any optimizations or modifications made based on simulation outcomes.

6. Implementation Result

Present the results of your implementation. Discuss:

- The performance metrics and evaluation criteria.
- Results obtained from testing and validation.
- Comparisons with existing solutions or benchmarks.
- Analysis and interpretation of results.
- Any unexpected findings or observations.

7. Conclusion

Summarize the key findings and contributions of your thesis. Include:

- A recap of the research problem and objectives.
- Main findings and their implications.
- Limitations of your research.
- Suggestions for future work and potential directions for further research.

References

[1] N. World, "IDC: Expect 175 zettabytes of Data Worldwide by 2025," accessed Dec. 1, 2023. [Online]. Available: `https://www.networkworld.com/article/966746/idc-expect-175-zettabytes-of-data-worldwide-by-2025.html`.

[2] J. Xu, S. Sarkar, H. Wang, and L. Hu, "Improving bounds on elliptic curve hidden number problem for ecdh key exchange," in *International Conference on the Theory and Application of Cryptology and Information Security*, Springer, 2022, pp. 771–799.